

```
1 C:\Users\13400f_RTX4060ti\AppData\Local\Programs\Python\
  Python313\python.exe D:\MoleColyte\Training_Scripts\train_Tox21.py
2 Splitting: Train: 4408 |Val: 551 | Test: 551
3 Splitting: Train: 103208 |Val: 12901 | Test: 12901
4 Kiln: cuda
5 Scanning training data to calculate exact imbalance penalties...
6 Calculated Pathway Penalties: [35.4 56.2 8.1 30.5 8.5 24.2 54.8 7.9 29.8
  22.7 7.7 19.8]
7
8 Starting Epoch 1
9   Epoch 1/20 | Batch 0 | Loss: 178.5766
10 ==> Epoch 1 Train Complete | Average BCE Loss: 48.5991
11   Validation Phase | Avg Val Loss: 5.7031
12   □ New best Tox21 model saved! (Lowest Val Loss: 5.7031)
13
14 Starting Epoch 2
15   Epoch 2/20 | Batch 0 | Loss: 12.2378
16 ==> Epoch 2 Train Complete | Average BCE Loss: 7.0567
17   Validation Phase | Avg Val Loss: 1.3992
18   □ New best Tox21 model saved! (Lowest Val Loss: 1.3992)
19
20 Starting Epoch 3
21   Epoch 3/20 | Batch 0 | Loss: 1.2662
22 ==> Epoch 3 Train Complete | Average BCE Loss: 1.3013
23   Validation Phase | Avg Val Loss: 1.4202
24   Model did not improve. Best Val loss remains: 1.3992
25
26 Starting Epoch 4
27   Epoch 4/20 | Batch 0 | Loss: 1.2591
28 ==> Epoch 4 Train Complete | Average BCE Loss: 1.3032
29   Validation Phase | Avg Val Loss: 1.4156
30   Model did not improve. Best Val loss remains: 1.3992
31
32 Starting Epoch 5
33   Epoch 5/20 | Batch 0 | Loss: 1.6543
34 ==> Epoch 5 Train Complete | Average BCE Loss: 1.3041
35   Validation Phase | Avg Val Loss: 1.4085
36   Model did not improve. Best Val loss remains: 1.3992
37
38 Starting Epoch 6
39   Epoch 6/20 | Batch 0 | Loss: 1.0115
40 ==> Epoch 6 Train Complete | Average BCE Loss: 1.3058
41   Validation Phase | Avg Val Loss: 1.4066
42   Model did not improve. Best Val loss remains: 1.3992
```

43
44 **Starting Epoch 7**
45 **Epoch 7/20 | Batch 0 | Loss: 1.2139**
46 **==> Epoch 7 Train Complete | Average BCE Loss: 1.3050**
47 **Validation Phase | Avg Val Loss: 1.4087**
48 **Model did not improve. Best Val loss remains: 1.3992**
49
50 **Starting Epoch 8**
51 **Epoch 8/20 | Batch 0 | Loss: 1.2695**
52 **==> Epoch 8 Train Complete | Average BCE Loss: 1.3042**
53 **Validation Phase | Avg Val Loss: 1.4173**
54 **Model did not improve. Best Val loss remains: 1.3992**
55
56 **Starting Epoch 9**
57 **Epoch 9/20 | Batch 0 | Loss: 1.1832**
58 **==> Epoch 9 Train Complete | Average BCE Loss: 1.3051**
59 **Validation Phase | Avg Val Loss: 1.4193**
60 **Model did not improve. Best Val loss remains: 1.3992**
61
62 **Starting Epoch 10**
63 **Epoch 10/20 | Batch 0 | Loss: 1.5703**
64 **==> Epoch 10 Train Complete | Average BCE Loss: 1.3071**
65 **Validation Phase | Avg Val Loss: 1.4186**
66 **Model did not improve. Best Val loss remains: 1.3992**
67
68 **Starting Epoch 11**
69 **Epoch 11/20 | Batch 0 | Loss: 1.6603**
70 **==> Epoch 11 Train Complete | Average BCE Loss: 1.3070**
71 **Validation Phase | Avg Val Loss: 1.4201**
72 **Model did not improve. Best Val loss remains: 1.3992**
73
74 **Starting Epoch 12**
75 **Epoch 12/20 | Batch 0 | Loss: 1.3161**
76 **==> Epoch 12 Train Complete | Average BCE Loss: 1.3060**
77 **Validation Phase | Avg Val Loss: 1.4181**
78 **Model did not improve. Best Val loss remains: 1.3992**
79
80 **Starting Epoch 13**
81 **Epoch 13/20 | Batch 0 | Loss: 1.1530**
82 **==> Epoch 13 Train Complete | Average BCE Loss: 1.3059**
83 **Validation Phase | Avg Val Loss: 1.4183**
84 **Model did not improve. Best Val loss remains: 1.3992**
85
86 **Starting Epoch 14**

```
87     Epoch 14/20 | Batch 0 | Loss: 1.1304
88 ==> Epoch 14 Train Complete | Average BCE Loss: 1.3055
89     Validation Phase | Avg Val Loss: 1.4179
90 Model did not improve. Best Val loss remains: 1.3992
91
92 Starting Epoch 15
93     Epoch 15/20 | Batch 0 | Loss: 0.9910
94 ==> Epoch 15 Train Complete | Average BCE Loss: 1.3068
95     Validation Phase | Avg Val Loss: 1.4181
96 Model did not improve. Best Val loss remains: 1.3992
97
98 Starting Epoch 16
99     Epoch 16/20 | Batch 0 | Loss: 1.4162
100 ==> Epoch 16 Train Complete | Average BCE Loss: 1.3063
101     Validation Phase | Avg Val Loss: 1.4202
102 Model did not improve. Best Val loss remains: 1.3992
103
104 Starting Epoch 17
105     Epoch 17/20 | Batch 0 | Loss: 1.0795
106 ==> Epoch 17 Train Complete | Average BCE Loss: 1.3057
107     Validation Phase | Avg Val Loss: 1.4198
108 Model did not improve. Best Val loss remains: 1.3992
109
110 Starting Epoch 18
111     Epoch 18/20 | Batch 0 | Loss: 1.5302
112 ==> Epoch 18 Train Complete | Average BCE Loss: 1.3058
113     Validation Phase | Avg Val Loss: 1.4194
114 Model did not improve. Best Val loss remains: 1.3992
115
116 Starting Epoch 19
117     Epoch 19/20 | Batch 0 | Loss: 1.3109
118 ==> Epoch 19 Train Complete | Average BCE Loss: 1.3057
119     Validation Phase | Avg Val Loss: 1.4188
120 Model did not improve. Best Val loss remains: 1.3992
121
122 Starting Epoch 20
123     Epoch 20/20 | Batch 0 | Loss: 1.3648
124 ==> Epoch 20 Train Complete | Average BCE Loss: 1.3058
125     Validation Phase | Avg Val Loss: 1.4185
126 Model did not improve. Best Val loss remains: 1.3992
127
128 Time: 0.30 minutes.
129
130 Process finished with exit code 0
```